

Task 1 – Mountain name entity recognition

Potential improvements

1. Dataset Expansion and Quality

The synthetic dataset generated by a single LLM (Gemini) exhibits limited structural variety and may contain systematic annotation bias. To address this, the following should be incorporated:

Real-world text corpora (e.g., Wikipedia articles, news outlets, travel blogs) to ensure better generalization;

Cross-validation of annotations using a secondary model or human annotators to establish inter-annotator agreement;

Hard negative samples and edge cases, such as ambiguous names (e.g., *Kilimanjaro* as a clothing brand), homonyms, or mountain names lacking explicit triggers like the word "Mount".

2. Transition to Larger Model Architectures

While DistilBERT serves as an efficient baseline, achieving higher precision requires scaling to larger architectures:

Full-scale models, such as BERT, RoBERTa, or domain-adapted variants (e.g., GeoBERT);

Standardized evaluation frameworks, specifically entity-level (rather than token-level) F1, precision, and recall metrics to rigorously justify model selection.

3. Edge Case Handling and Post-Processing

Rule-based and post-processing logic should be implemented for:

Multi-word and nested entities (differentiating mountain ranges from individual peaks);

Entity Resolution and Disambiguation, linking extracted entities to external knowledge bases (e.g., GeoNames);

Case sensitivity and abbreviation handling (e.g., *K2*, *Mt. Everest*).

4. Validation and Quality Assurance

Error Analysis, specifically targeting boundary confusion in entity extraction and false positives caused by lexical similarity.